Properties of the Fourier Series



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Key Concepts

1) The properties of the Fourier series are analogous to the properties of the other Fourier representations and are a result of the properties of sinusoids.

    a) Help manipulate and work with Fourier series.

    b) Provide insight and understanding into the nature of Fourier series representations.

2) Assumptions

$$x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k] \qquad y(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} Y[k]$$

where both $x(t)$ and $y(t)$ have fundamental period T and $\Omega_o = 2\pi/T$.

3) Linearity

$$ax(t) + by(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} aX[k] + bY[k]$$

4) Time shift

$$x(t - t_o) \overset{FS;\,\Omega_o}{\longleftrightarrow} e^{-jk\Omega_o t_o} X[k]$$

5) Frequency shift

$$e^{jk_o\Omega_o t} x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k - k_o]$$

6) Scaling

$$x(at) \overset{FS;\, a\Omega_o}{\longleftrightarrow} X[k]$$

7) Differentiation in time

$$\frac{d}{dt}x(t) \overset{FS;\, \Omega_o}{\longleftrightarrow} jk\Omega_o X[k]$$

8) Multiplication

$$x(t)y(t) \overset{FS;\, \Omega_o}{\longleftrightarrow} \sum_{p=-\infty}^{\infty} X[p]Y[k-p]$$

9) Parseval's Theorem

$$\frac{1}{T}\int_{<T>} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |X[k]|^2$$

10) Conjugate symmetry for $x(t)$ real valued

$$X[-k] = X^*[k]$$

which implies even symmetry for

$$|X[k]| = |X[-k]|, \quad \text{Re}\{X[k]\} = \text{Re}\{X[-k]\}$$

and odd symmetry for

$$\arg\{X[k]\} = -\arg\{X[-k]\}, \quad \text{Im}\{X[k]\} = -\text{Im}\{X[-k]\}$$

Video Screenshots

Properties of the Fourier series are similar to those of other Fourier representations - sinusoids

- Useful for manipulating Fourier series
- Develop insight and understanding

Assumptions

$$x(t) \overset{FS;\Omega_0}{\longleftrightarrow} X[k] \quad , \quad y(t) \overset{FS;\Omega_0}{\longleftrightarrow} Y[k]$$

Period T, $\Omega_0 = 2\pi/T$

1) Linearity

$$ax(t) + by(t) \overset{FS;\Omega_0}{\longleftrightarrow} aX[k] + bY[k]$$

2) Time shift

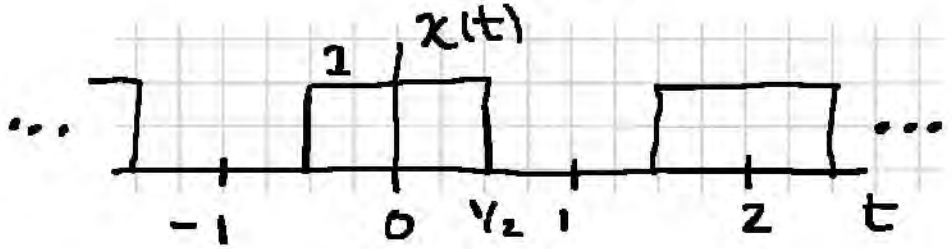
$$x(t-t_0) \xleftrightarrow{FS;\Omega_0} e^{-jk\Omega_0 t_0} X[k]$$

$$\text{if } x(t)=\sum_{k=-\infty}^{\infty} X[k]e^{jk\Omega_0 t} \Rightarrow x(t-t_0)=\sum_{k=-\infty}^{\infty} X[k]e^{jk\Omega_0(t-t_0)}$$

$$= \sum_{k=-\infty}^{\infty} \underbrace{X[k]e^{-jk\Omega_0 t_0}}\ e^{jk\Omega_0 t}$$

Example:

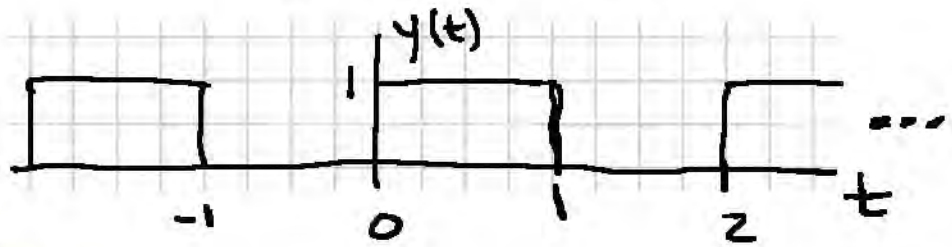
Given



$FS;\pi$

$$X[k] = \frac{\sin(k\pi/2)}{\pi k}$$

Find $Y[k]$



Solution: $y(t)=x(t-1/2)$

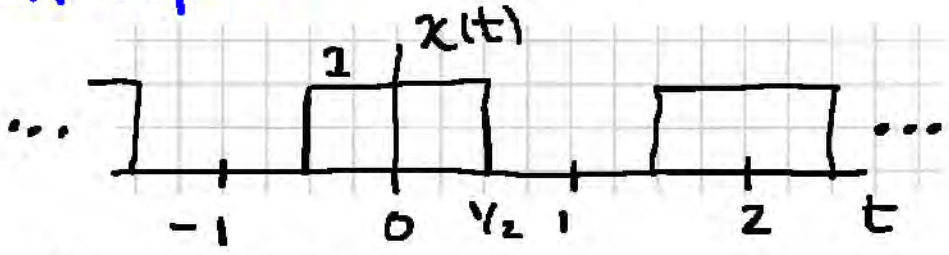
time-shift property

$$Y[k] = e^{-jk\pi/2}\ \frac{\sin(k\pi/2)}{\pi k}$$

3) Frequency Shift

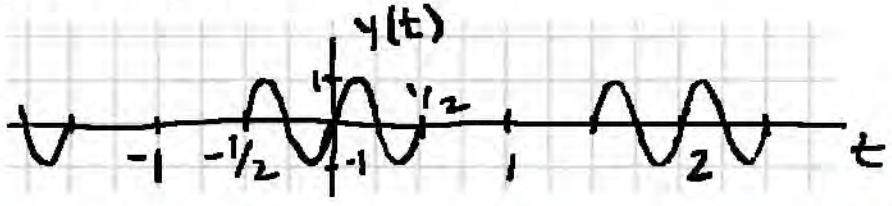
$$e^{jk_0\Omega_0 t}\,x(t) \xleftrightarrow{FS;\,\Omega_0} X[k-k_0]$$

Example: Given



$$\xleftrightarrow{FS;\,\pi} X[k] = \frac{\sin(k\pi/2)}{\pi k}$$

Find $Y[k]$



$$y(t) = \sin 4\pi t \;\; x(t)$$

Solution: $y(t) = \frac{1}{2j}e^{j4\pi t}x(t) - \frac{1}{2j}e^{-j4\pi t}x(t)$

by linearity + frequency shift $Y[k] = \frac{1}{2j}X[k-4] - \frac{1}{2j}X[k+4]$

4) Scaling

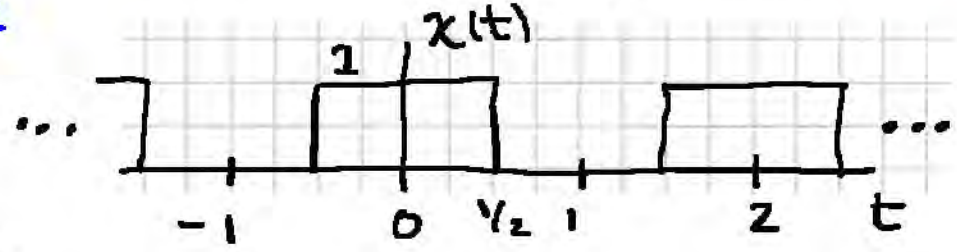
$$x(at) \xleftrightarrow{FS;\,a\Omega_0} X[k]$$

4) Differentiation in time

$\frac{d}{dt}x(t) \xleftrightarrow{FS_{j\Omega_0}} jk\Omega_0 X[k]$

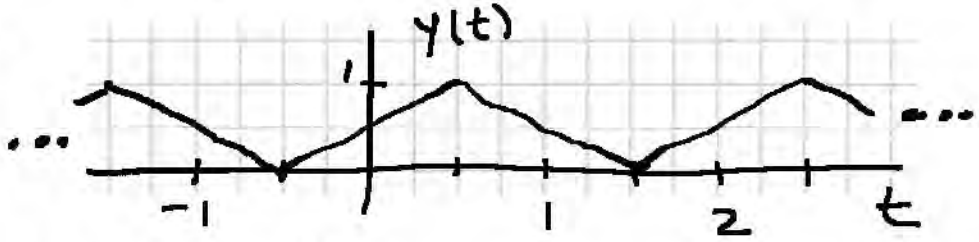
Example

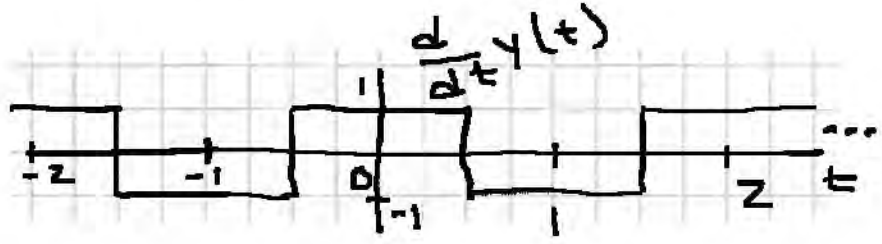
Given



$\xleftrightarrow{FS_{j\pi}}$ $X[k] = \frac{\sin(k\pi/2)}{\pi k}$

Find $Y[k]$





So $x(t) = \frac{1}{2}\left(\frac{d}{dt}y(t) + 1\right) \Rightarrow \frac{d}{dt}y(t) = 2x(t) - 1$

$jk\pi\, Y[k] = \frac{2\sin(k\pi/2)}{\pi k} - 1 \Rightarrow Y[k] = \frac{2\sin(k\pi/2)}{j\pi^2 k^2} - \frac{1}{jk\pi}, \quad k \neq 0$

$Y[0]$ is $\frac{1}{2}\int_{-1/2}^{3/2} y(t)\,dt = \frac{1}{2}$

5) Multiplication

$$x(t)\,y(t) \xleftrightarrow{FS;\,\Omega_0} \sum_{\ell=-\infty}^{\infty} X[\ell]\,Y[k-\ell]$$

"convolution"

6) Parseval's Theorem

$$\frac{1}{T}\int_0^T |x(t)|^2\,dt = \sum_{k=-\infty}^{\infty} |X[k]|^2$$

Example: $x(t) = \cos(4\pi t) \xleftrightarrow{FS;\,4\pi} X[k] = \begin{cases} 1/2, & k=-1,1 \\ 0, & \text{otherwise} \end{cases}$

Find $\int_0^{1/2} |x(t)|^2\,dt$.

Solution: $T = 1/2 \Rightarrow$

$$\frac{1}{1/2}\int_0^{1/2} |x(t)|^2\,dt = \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2 = 1/2$$

$$\text{So} \quad \int_0^{1/2} |x(t)|^2\,dt = (1/2)\cdot(1/2) = 1/4$$

7) Symmetry

$x(t)$ real $\xleftrightarrow{FS;\,\Omega_0}$ $X[-k] = X^*[k]$

$|X[k]| = |X[-k]|$ even

$\arg\{X[k]\} = -\arg\{X[-k]\}$ odd

$\mathrm{Re}\{X[k]\} = \mathrm{Re}\{X[-k]\}$ even

$\mathrm{Im}\{X[k]\} = -\mathrm{Im}\{X[-k]\}$ odd

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